

Contract-row frequencies, trigger explainability, basis risk

THE PUBLISHED QUESTION, CONDITIONAL ON FORMATION. If a cyclone were to form within **250 km** of **28.0°N 88.7°W** — the declared pre-genesis reference cell — in **August or September**, in seasons from **1971** onwards, what happened to storms that formed there? **SUFFICIENT**, N = 12, ESS 12, min 10.

OPERATIONAL AS OF **01 SEP 2026** · **16:08 CT / 21:08 UTC** CELL **28.0°N 88.7°W** · **R 250 KM** · **AUG–SEP** · **1971+**
PACK **134661125525F27A**

IT SAYS NOTHING ABOUT WHETHER THE SYSTEM FORMS. An unconditional intensity probability would require an external formation probability on the **same formation event and conditioning set**; none is computed here, and an NHC outlook probability is not multiplied by these rows unless the conditioning events are demonstrably aligned. **OPERATIONAL CONTEXT AS OF 01 SEP 2026 – HISTORICAL, NOT A CURRENT READING:** AL052026 (Tropical Storm Edouard, declared as Invest 97L) was at 29.9N 94.0W, 45 kt on NHC advisory 006; the archive holds no genesis row for it. **TD is definitional.** Status tokens: **OUT OF SCOPE -- unscoreable here; BASE RATE ONLY -- unscoreable.**

01 CONTRACT-RW FREQUENCIES			CONUS any 10/12 83.3% [55–95%] · CONUS ≥64 kt 1/12 8.3% [1–35%] · no state-level row exists		
INTENSITY · GENESIS-CONDITIONED			LANDFALL · SCORED REGIONS		LANDFALL · CONTINUED
reached TD	100.0%		CONUS — any	83.3%	CENTRAL AMERICA — any
12 / 12 · [76–100%]			10 / 12 · [55–95%]		0 / 12 · [0–24%]
reached TS	83.3%		CONUS — ≥64 kt	8.3%	CENTRAL AMERICA — ≥64 kt
10 / 12 · [55–95%]			1 / 12 · [1–35%]		0 / 12 · [0–24%]
reached Cat 1	25.0%		MEXICO — any	8.3%	HAWAII — any
3 / 12 · [9–53%]			1 / 12 · [1–35%]		0 / 12 · [0–24%]
reached Cat 2	16.7%		MEXICO — ≥64 kt	8.3%	HAWAII — ≥64 kt
2 / 12 · [5–45%]			1 / 12 · [1–35%]		0 / 12 · [0–24%]
reached Cat 3	16.7%		CARIBBEAN — any	0.0%	UNATTRIBUTED — any
2 / 12 · [5–45%]			0 / 12 · [0–24%]		0 / 12 · [0–24%]
reached Cat 4	8.3%		CARIBBEAN — ≥64 kt	0.0%	UNATTRIBUTED — ≥64 kt
1 / 12 · [1–35%]			0 / 12 · [0–24%]		0 / 12 · [0–24%]
reached Cat 5	8.3%				
1 / 12 · [1–35%]					

SAMPLE BOUNDARY 250 km · Aug–Sep · N = 12 SUFFICIENT → 150 km · N = 5 < min 10 RATE REFUSED → all months · N = 17 SUFFICIENT

02 TRIGGER EXPLAINABILITY, REFUSAL, NEAR MISSES

WHY A REPLAYABLE COHORT IS THE ARTEFACT

A parametric trigger's basis-risk file needs a prior a counterparty can reopen, not a number in a deck. The cohort is a URL: cell, radius, month window, season floor and methodology version travel in it, so the same query returns the same n/N and interval on any desk, and the denominator is named.

Cohort research, trigger design and counterparty explanation — not live pricing.

THE REFUSAL A COUNTERPARTY CAN CHECK

Tighten the same question from 250 km to **150 km** and the cohort falls to **N = 5**. Every rate then **REFUSES**: the engine returns “5 storms with a known outcome < min_sample=10” on every row. Counts still publish — CONUS any 4 of 5 — because a count is a fact and a rate over 5 storms is not. That refusal has its own replay URL below.

BASIS RISK – READ BEFORE USING ANY ROW ABOVE

- Same genesis cell ≠ same landfall.** Twelve storms formed inside one 250 km circle and went to Texas, Louisiana, Florida and Tamaulipas.
- Same class ≠ same local outcome.** The archive scores a crossing and its intensity, not a footprint or a loss.
- A CONUS row is not a state row.** No state-level landfall is scored, at any radius or sample size.
- The interval is the precision.** 1 of 12 is 8.3%, 95% Wilson 1-35%.

NEAR MISSES. **Anita 1977** formed 126 km from the cell and struck Tamaulipas at 120 kt – inside the cohort, outside the CONUS row. **Alicia 1983** formed 194 km out and struck Texas at 100 kt: the single member behind CONUS – ≥64 kt, and at 150 km not a member. **Barry 2019** formed 110 km out with a 65 kt Louisiana landfall, but its 11 July genesis is outside the window.

03 WHAT STORM ATLAS ADDS, AND THE COHORTS TO CITE

WHAT ATLAS ADDS. A public map gives position, P(forms) and a cone; Atlas gives exact n / N with a 95% Wilson interval on every contract row, a visible refusal where the record runs out, and a URL that reproduces both.

CITE THIS COHORT

REPLAY PUBLISHED COHORT · N=12

REPLAY 150 KM REFUSAL · N=5

REPLAY ALL MONTHS · N=17

STORM ATLAS · North Atlantic storms that formed within 250 km of 28.0°N 88.7°W, in August or September, in seasons from 1971 onwards, that went on to any outcome · 12 of 3,959 storms · METHODOLOGY 1.1.0 · PACK 134661125525f27a

250 KM · AUG–SEP <https://alecmessino.github.io/category-alpha/storm-atlas/?v=1&w=28%2C-88.7%2C250&s0=1971&mo=8.9&b=NA&m=1.1.0>

150 KM <https://alecmessino.github.io/category-alpha/storm-atlas/?v=1&w=28%2C-88.7%2C150&s0=1971&mo=8.9&b=NA&m=1.1.0>

ALL MONTHS <https://alecmessino.github.io/category-alpha/storm-atlas/?v=1&w=28%2C-88.7%2C250&s0=1971&b=NA&m=1.1.0>

RESEARCH ONLY – NOT A FORECAST. Every rate here is GENESIS-CONDITIONED: it assumes formation and describes what the record did next – not P(forms), not a live feed, not a cone. AN UNCONDITIONAL INTENSITY PROBABILITY REQUIRES AN EXTERNAL FORMATION PROBABILITY DEFINED ON THE SAME FORMATION EVENT AND CONDITIONING SET. None is computed here. An NHC outlook probability is not multiplied by these Atlas rows unless the conditioning events are demonstrably aligned. Landfall is counted jointly and is never decomposed into path/intensity marginals.

ARCHIVE 3,959 storms · 224,153 track points · 3,379 landfall rows · 33,054 environment records · METHODOLOGY 1.1.0 · PACK 134661125525f27a · BUILT 2026-08-18T05:18:57Z · SOURCES

ibtracs.EP.csv · ibtracs.NA.csv · ships.EP.txt · ships.CP.txt · ships.AL.txt · hurdat2.nepac.txt · hurdat2.atl.txt

RESEARCH ONLY

NOT A FORECAST

MILLIBAR / STORM ATLAS